



Theoretical study of the size confinement effect in linear π -conjugated oligomers

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Received 5 May 2007; accepted 28 September 2007

Available online 6 October 2007

Abstract

Linear π -conjugated oligomers represent model systems suitable for studying the size confinement effect in their spectroscopic response. In our study, based on Frenkel exciton models, we demonstrate that the excitonic approach successfully describes the experimentally observed size-dependent red shift of the optical spectra in thiophene- and benzene-based oligomer series. In addition, we will show that such an approach gives an interpretation of the excitation delocalization in such monodisperse linear conjugated oligomers in terms of the parameters of Frenkel exciton models. In our approach, these parameters are the oligomer length, on-site excitation energies, and the nearest-neighbor coupling strength. These parameters are obtained by fitting the curve, which represents the wave length value of the theoretical absorption line as a function of the oligomer length, to the experimental data. The definition of the structure of the repeat unit in the model description of oligomers is discussed.

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Keywords: Oligomers; Frenkel excitons; Conjugation length

1. Introduction

A strong interest in the photophysical properties of π -conjugated oligomers of definite length is evolving from the fact that these systems possess a monodisperse and perfectly defined chemical structure in contrast to polydisperse π -conjugated polymers. The dependence of the electronic properties on the chain length of conjugated oligomers has been widely studied [1–3]. Experimentally the advantage of the oligomeric approach can be seen in the resulting clear structure–property relationships in which typically, for shorter oligomers, a linear dependence of the electronic properties (absorption, emission energies, redox potentials) on the inverse chain length is found. The optical properties attributed to conjugated oligomers are directly related to their structure which typically means that the lowest ener-

getic absorption band has a π – π^* transition character which involves weakly bound π -electrons, determining the electronic properties of these oligomers. Typically, the π – π^* absorption band undergoes a progressive red shift with increasing length of the conjugated oligomer measured by the number of monomer repeat units. Several series of conjugated oligomers consisting of repeat units based on thiophenes, phenylenes or mixed building blocks (in the following also denoted as monomers) have been synthesized and their photophysical properties analyzed [4–13]. A general non-linear dependence of the absorption energy on the number of repeat units is observed for all these systems. Furthermore, saturation appears for the absorption energy of the oligomers, being more pronounced with increasing length of the oligomers. The possibility to synthesize systematically oligothiophenes [4–6] with controlled chain length even up to 96 repeat units [7] allows one to study size-dependent effects with high precision.

The problem of the size confinement effect in linear π -conjugated oligomers discussed by us already received a

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large attention and has a broad literature. In general conjugated oligomers and polymers are systems with three competing interactions: the electron bandwidth, the electron–phonon coupling energy, and the correlation energy. For a given geometry (the phonon variables are fixed) the model Hamiltonian for π -conjugated systems is typically called PPP (Pariser–Parr–Pople) and is widely discussed [14,15]. Such a Hamiltonian can be treated exactly (albeit, numerically) but for short chains only [16]. A different approach is based on quantum chemical calculations. The first extended study of the energy scheme and the developing band structure as a function of the oligomer length for oligothiophenes using time-dependent density-functional theory (TD-DFT) were presented by Telesca et al. [17]. The study of phenylene–acetylene and oligo(*p*-phenylenevinylene)s were presented by Magyar et al. [18] and Tretiak et al. [19]. Such studies in general reproduce a good tendency but typically experimental values cannot be fitted because usually estimates given by TD-DFT calculations are lower than expected. In the case of oligothiophenes the difference can be even around 1 eV [17]. This band-gap problem for longer chains is a common weakness of most density-functional based approaches although for PPV scaling and saturation of the experimental band-gap was accurately fitted [20].

Synthetic chemists developed a very useful fitting procedure [9] to study the limiting values of the π – π^* energies which for a great variety of different conjugated oligomers can be used. However, the suggested fitting procedure was used without an explanation of its physical or chemical validation. In a recent review paper [21] also a phenomenological fitting function was used for the description of optical properties of conjugated systems. In contrast in this study we suggest the use of fitting functions which were derived in the framework of microscopic models based on the Frenkel exciton picture and thus bring a physical insight into the material. Our approach is motivated by the fact that an excitonic picture for π -electrons in conjugated polymers is already well established in terms of transformations which lead the PPP Hamiltonian into an excitonic form [14,22,23] and by DFT results which clearly show that excitations in thiophene and phenylene-based oligomer chains can be well treated in the Frenkel exciton limit [17,18]. It was also confirmed that the linear optical absorption of quaterthiophene crystals is dominated by excitonic states [3]. In our previous study [11], we have shown that the linear oligothiophenes and ring shaped cyclothiophenes exhibit an interesting relation between their linear absorption spectra. In particular, there is a connection between the peak position of the oligothiophenes of given length and rings of a given size: the spectrum for the cyclothiophene with a given size corresponds to an oligothiophene chain of about half the length. As was shown [11], an approach based on the Frenkel exciton Hamiltonian, using simple geometry dependent selection rule arguments, nicely describes such an easily observed spectroscopic relationship between various cyclothiophenes and the corresponding oligothiophenes.

Our approach is based on the fact that in conjugated oligomers and polymers the effective strength of the Coulomb interaction leads to the formation of bound electron–hole pairs (excitons). The estimates of the exciton binding energy ranges from 0.2 to 0.9 eV [24,25] in such materials which is one to two orders of magnitude larger than typically observed in inorganic semiconductors. Such a large binding energy explains why at room temperature photophysics of conjugated systems can be dominated by the lowest energy exciton state. In our approach, we consider only the maximum of the absorption peak because the lowest energy exciton state, typically called 1^1B_u , carries most of the one photon intensity out of the electronic ground state. In addition, we would like to point out that the model discussed below cannot be taken as a general one for all conjugated systems. It is because only in the Frenkel limit the electron and the hole are located on the same building block of the model Hamiltonian. Over past many years it was discussed that in conjugated polymers the spatial electron–hole distance in excitons can extend over several unit cells, so in general they can be of charge-transfer or Wannier type [14]. The probability of finding an electron in a given building block when the hole is fixed in a particular building block can be calculated numerically and it was shown that these probability distributions vary strongly from polymer to polymer [20,25]. The two most extreme cases can be discussed by comparing the exciton features in polyacetylene (PA) and polyphenylenevinylene (PPV) [25]. In the PA case the electron distribution has a minimum at the hole position with a broad tail. In PPV the electron distribution is very narrow with a maximum at the hole position and consequently the Frenkel exciton limit is a good approximation for the bound electron–hole pair in PPV type polymers. One of the main goals of our study is to show that the most recent progress in the synthesis of the long thiophene and benzene based series nicely shows features which obviously can be explained within the Frenkel exciton model for these systems.

The paper is organized as follows: in the two following sections on Frenkel excitons a model with a single unit (interchangeably also denoted as building block or monomer) per unit cell will be considered first. The microscopic model parameters like on-site excitation energy and the nearest-neighbor coupling strength will be discussed in detail. In the Section 3 a model with two-building blocks per unit cell is investigated and the curve describing the excitation energy as a function of the number of building blocks is derived. Furthermore we discuss the connection with the approach of Section 2. A discussion and concluding remarks will be given at the end of this paper as Sections 4 and 5.

2. Excitation energy for a model with one-building block per repeat unit

To model the photophysical properties of oligothiophenes we consider a linear chain of N_1 (the index “1” indi-

cates the consideration of an open linear chain in contrast to possible cyclic chains [11]) optically active two-level monomers (in the sense of the last paragraph of Section 1) forming a simple one-dimensional open linear chain. As examples for the systems considered in this section, see **OT** and **Bu-OT** in Scheme 1, one thiophene ring (with six π -electrons) is approximated as a two-level monomer and in the case of **Bu-OT** the bare thiophene ring and the alkyl-substituted thiophene ring in the same repeat unit are approximated by two identical two-level monomers. In other words, as a first approximation in this theoretical section we consider linear chains with repeating identical monomer units only. Each monomer can be locally excited and the superposition of the local excitation of these monomers gives rise to collective excitations, i.e. to Frenkel excitons. In our study we will consider a single excitation only, therefore $\sum_{m=1}^{N_1} a_m^\dagger a_m = 1$, and the corresponding Frenkel exciton model Hamiltonian reads

$$H_{\text{ex}} = \omega_0 - J \sum_{m=1}^{N_1-1} (a_m^\dagger a_{m+1} + a_{m+1}^\dagger a_m), \quad (1)$$

where ω_0 is the excitation energy of each monomer, and $|m\rangle = a_m^\dagger |0\rangle$ denotes the excited state of the m th monomer. J describes the nearest-neighbor exciton transfer matrix element and the sign of J depends on the angle between the optical transition dipole moments and the chain axis. In our calculations we assume J to be *positive* and thus the states coupling to the electromagnetic wave are those

at the bottom of the exciton band (note the negative sign in the Hamiltonian). The Hamiltonian (1) can be diagonalized using the new basis $|\psi_k\rangle = \sum_m \phi_m(k) |m\rangle$ and for an open chain of length N_1 the eigenstates are given by

$$|\psi_k\rangle = \left(\frac{2}{N_1+1}\right)^{1/2} \sum_{m=1}^{N_1} \sin\left(\frac{mk\pi}{N_1+1}\right) |m\rangle, \quad (2)$$

with the corresponding eigenvalues

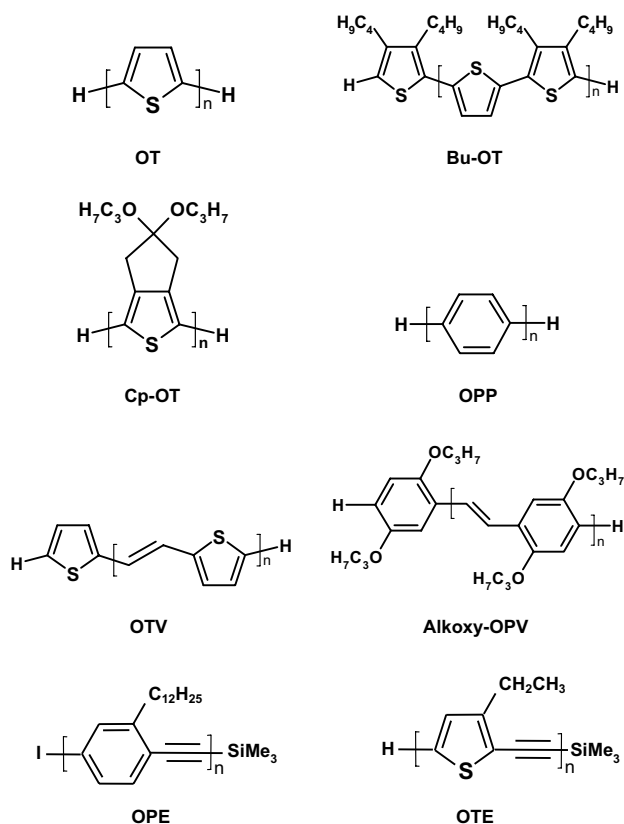
$$E_k = \omega_0 - 2J \cos\left(\frac{k\pi}{N_1+1}\right). \quad (3)$$

States with $k = 1, 2, \dots, N_1$ form an exciton band. The state with $k = 1$ lies at the bottom of the exciton band. In this simple picture, the optical absorption spectrum shows a series of peaks at the exciton eigenfrequencies, with weights given by the oscillator strengths $O_k = |\sum_m \phi_m(k) \mu_m|^2$ (μ_m is the transition dipole moment at site m , the polarization vector of the light and the average over all orientations of the oligomers are omitted). For an open chain, with transition dipoles in the monomer units equal in magnitude and parallel to each other, $\mu_m = \mu$, the oscillator strength takes the form

$$O_k^l = \frac{1 - (-1)^k}{2} \frac{2\mu^2}{N_1+1} \cot^2 \frac{k\pi}{2(N_1+1)}. \quad (4)$$

In this case only states with k odd have oscillator strength, while states with k even are dark. Most of the oscillator strength is accumulated in the lowest state ($O_{k=1} = 0.81(N_1+1)\mu^2$ for a long chain). Consequently, we can assume that the absorption maxima of the linear oligothiophenes are dominated by the optically allowed transition to the lowest state ($k = 1$) in the excitonic band.

Fig. 1, left panel, shows the maxima of the experimental absorption energies (squares) for the bare, non-substituted oligothiophenes series, **OT** (Scheme 1), in solution at room temperature. Using Eq. (3) with $k = 1$, for the model parameters $\omega_0 = 5.65$ eV and $J = 1.54$ eV a good fit is obtained (dashed line) with $R^2 = 0.9987$. (R^2 is a measure of the quality of the fit according to the Origin program and $R^2 = 1$ means perfect fit). Fig. 1 presents also an analysis of the data (dots) for the series of butyl-substituted oligothiophenes, **Bu-OT** (Scheme 1). The full line represents the fitted curve obtained from Eq. (3) for this system. The right panel shows the same data presented as a function of the inverse chain length $1/N_1$ to better depict the saturation for longer chains. Because in our model the delocalization length of the Frenkel exciton is assumed to be identical with the oligomer chain length, the observed saturation of the absorption maxima is determined by the chain length dependence of the lowest excitation energy. (If the delocalization length would be a monotonous function of the chain length, one would expect similar results. This would not be the case if the delocalization length would be shorter and independent of the chain length, see also the discussion in the following paragraph.) Although both oligomer series are based on the same



Scheme 1. Structures of the various oligomer series.

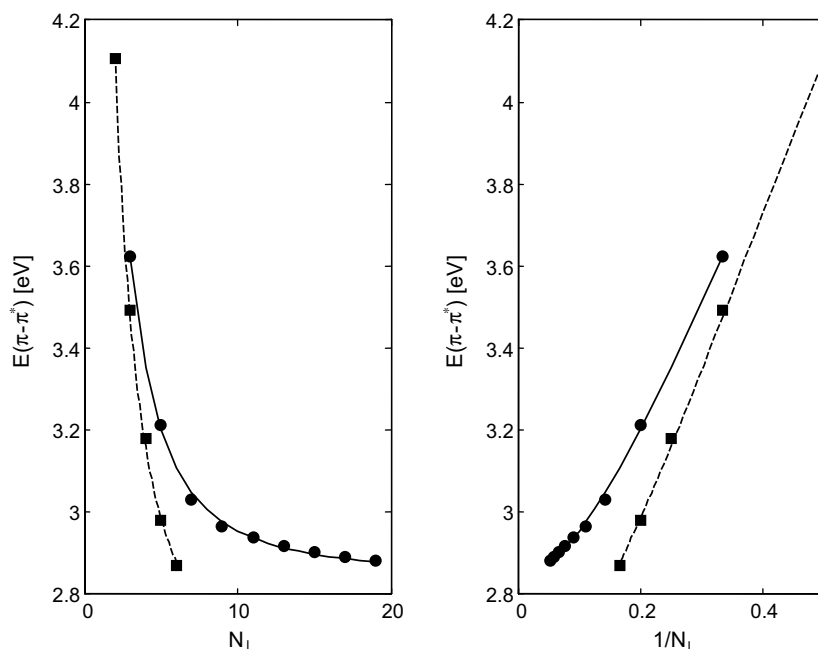


Fig. 1. The left panel shows the dependence of the maximum of the π - π^* absorption band on the size of the oligothiophenes **OT** and **Bu-OT**. The dots and squares represent the experimental data taken from the literature [9,11], while the solid and dashed lines represents the theoretical fit based on Eq. (3) ($k = 1$). The dashed line with $\omega_0 = 5.65$ eV and $J = 1.54$ eV, $R^2 = 0.9987$, represents the case of thiophenes without any side groups **OT**, see Scheme 1. The solid line with $\omega_0 = 5.5$ eV and $J = 1.33$ eV, $R^2 = 0.9988$, represents the case of **Bu-OT**. The right panel shows the same data presented as a function of the inverse chain length $1/N_1$ to better show saturation for longer chains.

aromatic unit, we can see that the solid and dashed curves do not overlap. It means that, as we can expect, the side groups attached at the β -position modify the electronic properties of the bare thiophene monomers. Therefore the fitted parameters obtained for **Bu-OT** are different from those of **OT** and given by $\omega_0 = 5.5$ eV and $J = 1.33$ eV.

The longest oligothiophene synthesized until now corresponds to cyclopentane-substituted oligothiophenes (**Cp-OT**) (Scheme 1) [7]. Fig. 2 shows the dependence of the maximum of the π - π^* absorption band on the length of the **Cp-OTs**. The experimental data are depicted as dots, while the solid line represents the theoretical curve based on Eq. (3) ($k = 1$) with $\omega_0 = 5.61$ eV and $J = 1.63$ eV. The inset shows the data presented as a function of the inverse chain length $1/N_1$. The fit confirms our supposition that the conjugation in these very long oligomers is not broken and furthermore is given by the oligomer length. As above, in our approach we assume that the delocalization length of the excitation, or conjugation length, is equal to the oligomer length. Indeed, our results seem to support this assumption because the fit, even for these very long systems, describes their electronic properties correctly. If in such very long oligomers the conjugation and therefore the excitation delocalization would be broken even only once, immediately this would show up as a much faster saturation of the peak positions than the experimentally observed one. In Fig. 2 it can be seen, that even the 96-mer shows a small red shift (1 nm) in comparison to the 72-mer. This is in accordance with Eq. (3), which shows that the larger N_1 the smaller is the shift of the absorption spectrum. All

experimental points in Fig. 2 are fitted very nicely by the fitting function (3), showing a large delocalization length in these impressively long oligomers. The higher value of the coupling constant ($J = 1.63$ eV vs. 1.54 eV for the bare oligothiophenes) describes a stronger coupling between the molecules and supports our conclusions.

Fig. 2, right panel, shows that our approach is not limited to thiophene-based oligomers. A more general demonstration of the usefulness of the expression given by Eq. (3) can be shown for the series of oligo(*p*-phenylenes), **OPP** (Scheme 1). The right panel of Fig. 2 shows the dependence of the maximum of the π - π^* absorption band on the size of **OPP**. The dots show the experimental data [10], while the solid line represents the theoretical curve based on Eq. (3) ($k = 1$) with $\omega_0 = 6.164$ eV and $J = 1.265$ eV. The inset panel shows the same data presented as a function of the inverse chain length $1/N_1$.

Although our approach describes well the experimental features of the two series of oligothiophenes described by now, we obviously have to discuss in which way the unit cell has to be modeled in the case of the mixed, substituted and non-substituted, oligothiophenes **Bu-OT**. In fact, we have two slightly different alternating units, one bare thiophene and one alkyl-substituted thiophene. Because we can intuitively assume that alkyl chains can induce some effects on the HOMO–LUMO gap of the thiophene unit, a more detailed model is investigated. Thus a modified model Hamiltonian taking into account the coexistence of two different building blocks in the unit cell will be developed in the next subsection.

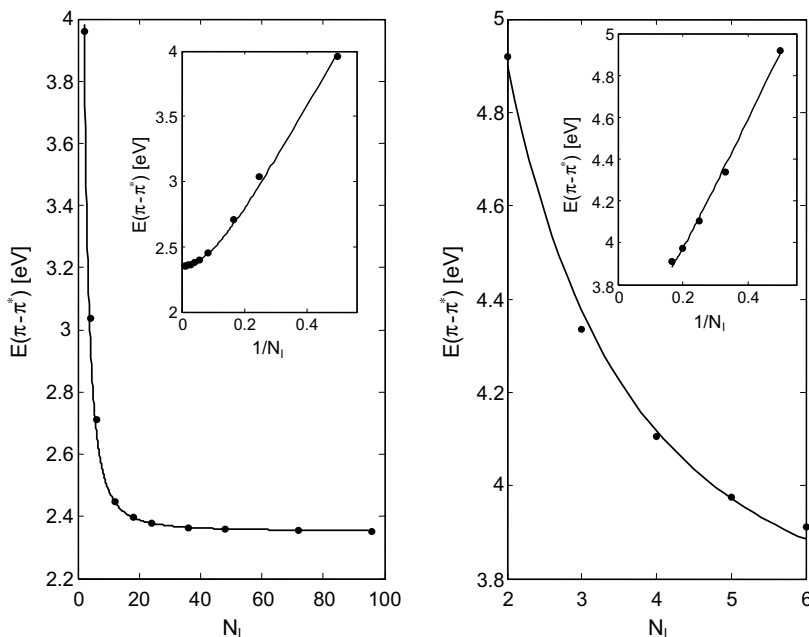


Fig. 2. The figure shows the dependence of the maximum of the π - π^* absorption band on the size of the cyclopentane-substituted oligothiophenes (**Cp-OT**, see Scheme 1) up to the 96-mer. The dots mark the experimental data taken from the literature [7], while the solid line connecting the dots represents the theoretical curve based on Eq. (3) ($k = 1$) with $\omega_0 = 5.61$ eV and $J = 1.63$ eV, $R^2 = 0.9979$. The inset shows the same data presented as a function of the inverse chain length $1/N_1$. The right panel shows the dependence of the maximum of the π - π^* absorption band on the size of the oligo(*p*-phenylenes) series, **OPP**. The dots mark the experimental data taken from the literature [10], while the solid line connecting the dots represents theoretical curves based on Eq. (3) ($k = 1$) with $\omega_0 = 6.164$ eV and $J = 1.265$ eV, $R^2 = 0.9957$. The inset panel shows the same data presented as a function of the inverse chain length $1/N_1$.

3. Excitation energy for a model with two different building blocks per repeat unit

In this section, we will discuss the case where one repeat unit is modeled by two different building blocks, building block₁ and building block₂, respectively. In the simplest case a building block is identical with what is usually called a monomer (see, e.g., **Bu-OT** in Scheme 1). In this case we could also talk about a chain with two monomers in the repeat unit. However we also allow for the situation, where the monomer can be considered as composed of two-building blocks (see, e.g., **OTV**, or **OTE** in Scheme 1). In the second case the repeat unit contains one monomer, but two-building blocks. In general the oligomer chain length N_1 (number of monomers) can be even or odd, in the latter case one of the monomers is unpaired; both cases will be discussed in one frame. The starting model Hamiltonian is similar to (1), with the only modification that instead of ω_0 at each site we have now a chain of alternating ω_1 and ω_2 values, which denote the on-site excitation energies of the two different monomers within the unit cell. For simplicity we assume that only one type of interaction between monomers exists and J can be kept as the only interaction parameter. In order to get a simple notation we will make the variable substitution $\omega_1 = \Delta + \delta$ and $\omega_2 = \Delta - \delta$ in the Hamiltonian (1). In this notation, the corresponding Frenkel exciton Hamiltonian reads

$$H_{\text{ex}} = \Delta + \sum_{m=1}^{N_1} (-1)^{m-1} \delta a_m^\dagger a_m - J \sum_{m=1}^{N_1-1} (a_m^\dagger a_{m+1} + a_{m+1}^\dagger a_m). \quad (5)$$

N_1 again counts the number of monomers or building blocks (and not the number of unit cells). The trivial part of the Hamiltonian (5) is the Δ term which can be excluded from the diagonalization since its value gives only a shift of the excitonic band. In this notation, we have again to diagonalize a symmetric tridiagonal matrix like (1) with the only difference that the diagonal elements have the same absolute values but with the sign in front of δ alternating. In the first step, we determine the eigenstates $|\psi\rangle$ and the energy spectrum of the open chain with two monomers per unit cell, with different local excitation energies, solving the stationary Schrödinger equation $H_{\text{ex}}|\psi_k\rangle = E_k|\psi_k\rangle$. Inserting $|\psi_k\rangle = \sum_m \varphi_m(k) a_m^\dagger |0\rangle$ into the Schrödinger equation we obtain the following set of equations for the expansion coefficients $\varphi_m(k)$

$$(E_k - (-1)^{m-1} \delta) \varphi_m(k) = J(\varphi_{m-1}(k) + \varphi_{m+1}(k)), \quad (6)$$

which are valid for $1 < m < N_1$. At first, let us select from (6) the three equations for $m = 2n$, $m = 2n + 1$, and $m = 2n - 1$, not presented here. After straightforward algebra we arrive at the following equation

$$(E_k^2 - \delta^2 - 2J^2) \varphi_{2n}(k) = J^2(\varphi_{2(n-1)}(k) + \varphi_{2(n+1)}(k)). \quad (7)$$

The last equation can be solved using the following ansatz, which is standard for such a type of equations (see e.g. [26,27] and Eq. (2) in the previous subsection),

$$\varphi_n(k) = C_{\text{even}}(k) \sin \frac{nk\pi}{N_1 + 1}, \quad (8)$$

where $C_{\text{even}}(k)$ is a real k -dependent constant, and the k numerate states in the band. Here, $k = 1, 2, \dots, N_1/2$, for N_1 even. With the help of simple trigonometric identities the secular equation takes the form

$$E_k^2 = \delta^2 + 2J^2 \left(1 + \cos \frac{2k\pi}{N_1 + 1} \right) \quad (9)$$

and the corresponding eigenvalues are just the square root of the last equation. For the starting Hamiltonian (5) of an open chain with two monomers per unit cell, the eigenvalues are obtained by adding the average excitation energy term, thus

$$E_k^{\text{tot}} = \Delta \pm \sqrt{\delta^2 + 4J^2 \cos^2 \frac{k\pi}{N_1 + 1}}. \quad (10)$$

This model leads to the occurrence of two separate single-exciton bands, one with the plus sign (upper band) and one with the minus sign (lower band) in Eq. (10). Such bands are named Davydov components [27]. In each band k numerates the states. Obviously, Eq. (10) reduces immediately to Eq. (3) in the case of $\omega_1 = \omega_2 = \omega_0$. Furthermore, for $N_1 = 2$ and $k = 1$, the last equation leads to the well known eigenvalue solution for the dimer problem with different on-site excitation energies, see e.g. [26]. It is important to realize that we get eigenvalues without solving all equations for all expansion coefficients ϕ_m . In fact up to now, we have got only one half of them, for $m = 2n$, see Eq. (7). For the case with m odd, we have equations similar to Eq. (7), and to find the rest of the missing expansion coefficients we can use the same ansatz as Eq. (8) but with different amplitudes $C_{\text{odd}}(k)$. It means that the wave functions of the Hamiltonian (5) are built from two different sets of coefficients, one for the odd sites in the chain and another one for the even sites, both described by the same sine function like in the previous case, see Eq. (2), but with different k -dependent amplitudes.

From the stationary Schrödinger equation we get for $m = 1$ and $m = N_1$ two additional equations which also have to be satisfied by our ansatz. In terms of our ansatz the extra equation for $m = 1$ reads $(E_k - \delta)C_{\text{odd}}(k) = 2JC_{\text{even}}(k) \cos \frac{k\pi}{1+N_1}$. In addition, with our ansatz we find that for $m = N_1$ with N_1 odd the extra equation exactly coincides with the one for $m = 1$. When N_1 is even we have an equation $(E_k + \delta)C_{\text{even}}(k) = 2JC_{\text{odd}}(k) \cos \frac{k\pi}{1+N_1}$, but it is easy to show with help of Eq. (9) that both equations are equivalent; $C_{\text{odd}}(k)$ and $C_{\text{even}}(k)$ satisfy both of them at the same time. Thus, independently of whether N_1 is odd or even, the amplitudes $C_{\text{odd}}(k)$ and $C_{\text{even}}(k)$ can be found from the following set of equations

$$\begin{cases} C_{\text{odd}}(k)^2 + C_{\text{even}}(k)^2 = \frac{4}{1+N_1} \\ (E_k - \delta)C_{\text{odd}}(k) = 2JC_{\text{even}}(k) \cos \frac{k\pi}{1+N_1} \end{cases} \quad (11)$$

The first equation comes from the normalization condition of the wavefunctions. From the solution of this simple set of equation it is seen that the amplitudes $C_{\text{odd}}(k)$ and $C_{\text{even}}(k)$ depend on J and δ through E_k (but not through E_k^{tot} , i.e. Δ does not influence the oscillator strength). However, the oscillator strength will also depend on J , which is in contrast to the case of a single monomer per unit cell. Nevertheless, also for odd values of N_1 all wave functions, like in the previous case, must be symmetrical or antisymmetrical with respect to the middle of the chain. Therefore for the assumed transition dipole orientation, differences in terms of oscillator strength distributions between the cases of one-building block per unit cell, discussed in the previous section, and of two-building blocks per unit cell must be quite small. As in the previous section again we have dark states, the even ones, and states which carry the oscillator strength, the odd states. As in the previous case, the bottom state as the most symmetric one without any nodes carries the largest oscillator strength and therefore these two cases can be compared very easily. Assuming, as in the previous section, that all transition dipoles are equal in magnitude and parallel to each other, the oscillator strength for N_1 odd takes the form

$$O_k = \frac{1 - (-1)^k}{2} \mu^2 \csc^2 \frac{k\pi}{1+N_1} \times \frac{k\pi}{1+N_1} \left(C_{\text{odd}}(k) + C_{\text{even}}(k) \cos \frac{k\pi}{1+N_1} \right)^2. \quad (12)$$

The last formula shows that because of symmetry the state for $k = 1$ takes almost all the oscillator strength (for allowed k values, the function $\csc^2 \frac{k\pi}{1+N_1}$ ($\csc x = 1/\sin x$) is quickly decreasing with k increasing). It is worth to mention that to properly calculate the oscillator strength according to Eq. (12) we need to care about the signs of $C_{\text{odd}}(k)$, $C_{\text{even}}(k)$, and E_k (E_k for the upper band changes the sign and $C_{\text{odd}}(k)C_{\text{even}}(k)/|C_{\text{odd}}(k)C_{\text{even}}(k)|$ is equal to 1 or -1 depending on the band). For completeness, we must add that in the case of N_1 odd, only $N_1 - 1$ eigenvalues of the Hamiltonian (5) are described by Eq. (10) (see the k values which label states in the bands). The last missing state is $\varphi_m = \sqrt{\frac{2}{N_1+1}} \sin \frac{mk\pi}{N_1+1}$ with $k = \frac{N_1+1}{2}$ and $E_k = \Delta + \delta = \omega_1$. The Davydov components are non-symmetric in this case. This state has nodes at all sites with excitation energy ω_2 and therefore this state remains unchanged and its eigenvalue does not depend on ω_2 . For $(1 + N_1)/2$ even this state is dark and when $(1 + N_1)/2$ is odd the oscillator strength of these states is simply given by $2/(1 + N_1)$. In the case of N_1 even the oscillator strength formula (12) does not apply. Lack of symmetry, with respect to the middle site of the chain, leads to the fact that now all states carry a nonzero oscillator strength and this can be easily seen even in the case of an ordinary dimer with different on-site energies where upper and lower states are bright, and

$O_{1,2} = \mu^2(1 \mp J/\sqrt{(\omega_1 - \omega_2)^2/4 + J^2})$. For longer chains the bottom state still contributes the dominant part to the total oscillator strength and Eq. (10) can be used as a fitting function in the same manner as in the case of N_1 odd.

4. Main results and discussion

In Sections 2 and 3 we have derived two fitting functions from two model Hamiltonians; one describing oligomers with one-building block per unit cell, given by Eq. (3) and the second one describing oligomers containing two-building blocks per unit cell, given by Eq. (10). With this second approach, we can now describe more precisely the substituted oligothiophene series, **Bu-OT**, and compare the parameters with those obtained from the approach given by Eq. (3) in Section 2 of this paper. The perfect fit based on Eq. (10) ($k = 1$, and lower Davydov component) is obtained with the following parameters: $\omega_1 = 5.65$ eV, $\omega_2 = 5.3$ eV and $J = 1.31$ eV. The resulting fitting curve is not shown because it overlaps exactly with the solid line already described in Fig. 1 supporting the consistency of both approaches.

A more general demonstration of the usefulness of the fitting formula given by Eq. (10) can be given for the series of thienylenevinylenes, **OTV** (Scheme 1). Here the two-building blocks, vinylene (double bond unit) and thiophene, differ not only energetically but also in their nature. The application of the non-linear least square (NLLS) fit-

ting procedure on Eq. (10) with three independent parameters (δ , A , and J) did not converge to a unique solution with small errors for all series because the number of experimental points was not large enough. In the case of the **OTV** series we already know the thiophene's HOMO–LUMO excitation $\omega_1 = 5.65$ eV from our **OT** estimate in the former section, see Fig. 1, and applying Eq. (10) with then only two independent fitting parameters (ω_2, J) we get a good estimate of the HOMO–LUMO transition for the vinylene unit. Fig. 3, left panel, shows the dependence of the maximum of the π – π^* absorption band on the size of the **OTVs** (dots) [8]. The solid line represents the theoretical curve based on Eq. (10) ($k = 1$, and “-”) with $\omega_1 = 5.65$ eV, $\omega_2 = 10.82$ eV (vinylene unit), and $J = 2.69$ eV. The inset shows the same data but presented as a function of the inverse chain length $1/N_1$. As expected the vinylene HOMO–LUMO transition energy is much larger than the thiophene one and the coupling constant increases considerably.

The possibility to generalize the second approach to other oligomer classes can be demonstrated for non-thiophene based oligomers. The right panel in Fig. 3 shows the dependence of the maximum of the π – π^* absorption band (dots) on the size of the dialkoxy-substituted oligophenylenevinylenes, **Alkoxy-OPV** (Scheme 1) [9]. The repeat unit is formed by two blocks: the benzene ring with attached side chains and the vinylene unit. The benzene-based compounds exhibit an excitonic behaviour as well

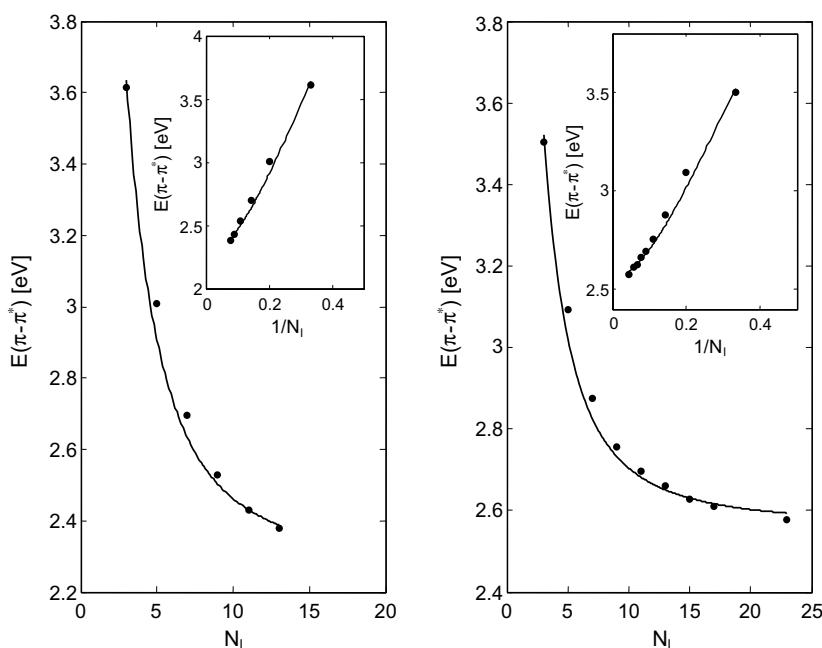


Fig. 3. The left panel shows the dependence of the maximum of the π – π^* absorption band on the size of the thienylenevinylene series (**OTV**). The dots mark the experimental data taken from the literature [8], while the solid line connecting the dots represents the theoretical curve based on Eq. (10) ($k = 1$, and “-”) with $\omega_1 = 5.65$ eV (thiophene chromophor), $\omega_2 = 10.82$ eV (vinylene chromophor), and $J = 2.69$ eV, $R^2 = 0.9980$. The inset panel shows the same data presented as a function of the inverse chain length $1/N_1$. The right panel shows the dependence of the maximum of the π – π^* absorption band on the size of the dialkoxy-substituted oligophenylenevinylenes, **Alkoxy-OPV**. The dots mark the experimental data taken from the literature [9], while the solid line connecting the dots represents theoretical curves based on Eq. (10) ($k = 1$, and “-”) with $\omega_1 = 4.77$ eV (benzene monomer), $\omega_2 = 10.82$ eV (vinylene monomer), and $J = 2.134$ eV, $R^2 = 0.9893$. The inset panel shows the same data presented as a function of the inverse chain length $1/N_1$.

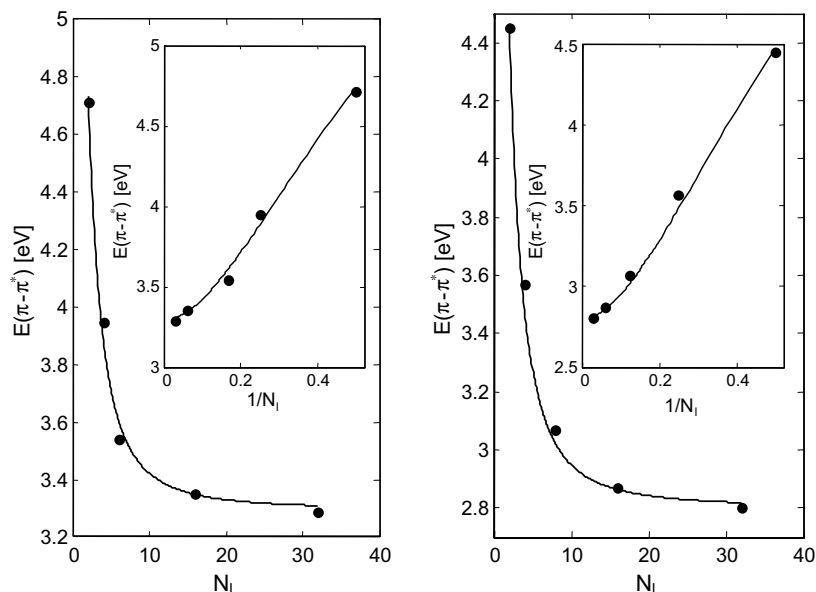


Fig. 4. The left panel shows the dependence of the maximum of the $\pi-\pi^*$ absorption band on the size of oligo(phenylene ethynylene)s, **OPE**. The dots mark the experimental data taken from the literature [12], while the solid line connecting the dots represents theoretical curves based on Eq. (10) ($k = 1$, and “-”) with $\omega_1 = 5.43$ eV (benzene monomer), $\omega_2 = 9.5$ eV (ethynylene monomer), and $J = 1.82$ eV, $R^2 = 0.9939$. The inset panel shows the same data but presented as a function of the inverse chain length $1/N_1$. The right panel shows the dependence of the maximum of the $\pi-\pi^*$ absorption band on the size of the oligo(thiophene ethynylene)s series, **OTE**. The dots mark the experimental data taken from the literature [13], while the solid line connecting the dots represents theoretical curves based on Eq. (10) ($k = 1$, and “-”) with $\omega_1 = 5.3$ eV (thiophene monomer), $\omega_2 = 9.5$ eV (ethynylene monomer), and $J = 2.05$ eV, $R^2 = 0.9959$. The inset panel shows the same data but presented as a function of the inverse chain length $1/N_1$.

[28]. In this case we will apply Eq. (10) taking the excitation energy of the vinylene unit fixed as 10.82 eV, known from our previous estimate. The solid line represents the theoretical curve based on Eq. (10) ($k = 1$, and “-”) with $\omega_1 = 4.77$ eV (benzene unit), $\omega_2 = 10.82$ eV (vinylene unit), and $J = 2.134$ eV. The inset panel shows the same data presented as a function of the inverse chain length $1/N_1$.

Another important group in benzene and thiophene based oligomers is represented by oligo(phenylene ethynylene)s (**OPE**) and oligo(thiophene ethynylene)s (**OTE**), see Scheme 1. In Fig. 4, the left panel shows the dependence of the maximum of the $\pi-\pi^*$ absorption band on the oligomer size in the **OPE** series. The dots mark the experimental data taken from the literature [12], while the solid line connecting the dots represents theoretical curves based on Eq. (10) ($k = 1$, and “-”) with $\omega_1 = 5.43$ eV (benzene monomer), $\omega_2 = 9.5$ eV (ethynylene monomer), and $J = 1.82$ eV. The inset panel shows the same data but presented as a function of the inverse chain length $1/N_1$. The right panel on Fig. 4 shows the dependence of the maximum of the $\pi-\pi^*$ absorption band on the size of the **OTE** series. The dots mark the experimental data taken from the literature [13] while the solid line connecting the dots represents the theoretical curve based on Eq. (10) ($k = 1$, and “-”) with $\omega_1 = 5.3$ eV (thiophene monomer), $\omega_2 = 9.5$ eV (ethynylene monomer), and $J = 2.05$ eV. The inset panel shows the same data but presented as a function of the inverse chain length $1/N_1$. Such accurate fits support the assumption that a double- or triple-bond unit attached to the thiophene or benzene aromatic ring can be treated as

a separate term in the model Hamiltonian. The parameters obtained from the fit to the **OPE** and **OTE** series are in good agreement with estimates given by the earlier considered oligomers series, namely that the on-site excitations for thiophene and benzene monomers with attached side groups have smaller values than for the bare ones. The estimate of the excitation in the triple bond unit is again quite high, like for the double bond unit, and equal to 9.5 eV.

All results are summarized in Table 1, where the upper part gives the parameters for the one-building block Hamiltonian resulting in Eq. (3) for the excitation energy and the lower part those for the two-building block Hamiltonian resulting in Eq. (10) for the excitation energy. The data allow for a qualitative comparative discussion with a well understandable physical background which takes into account the influence of the side chains and of the steric conditions on the excitation energy of the building block and the transfer integrals. For the bare thiophene

Table 1
Fitting parameters for the considered oligomer series based on Eq. (3) (upper part) and Eq. (10) (lower part)

Oligomer	OT	Bu-OT	Cp-OT	OPP	OTV	Alkoxy-OPV	OPE	OTE
ω_0	5.65	5.5	5.61	6.16	5.08	4.49	4.7	4.44
J	1.54	1.33	1.63	1.27	1.47	0.99	0.72	0.84
ω_1		5.65			5.65	4.77	5.43	5.3
ω_2		5.3			10.82	10.82	9.5	9.5
J		1.31			2.69	2.13	1.82	2.05

All presented values are given in eV.

oligomers, **OT**, the value of $\omega_0 = 5.65$ eV describes perfectly the absorption maximum for the monothiophene chromophore (220 nm). The introduction of alkyl-substituents, **Bu-OT**, leads to a red shift in the excitation energy of the substituted building block two ($\omega_2 = 5.3$ eV from Eq. (10)) due to the electron donating effect of the alkyl chains to the π -system. The reduced coupling constant J (interaction parameter of the two-building block exciton model) for the substituted oligomers, **Bu-OT** (1.31 eV vs. 1.54 eV for **OT**), can be explained by the increased torsion angle between the thiophenes caused by the alkyl chains. The loss of planarity is a well-known consequence of β -alkylation in thiophenes. Comparing the results from both, Eq. (3) ($\omega_0 = 5.5$ eV) and Eq. (10) ($\omega_1 = 5.65$ eV, $\omega_2 = 5.3$ eV) for the **Bu-OT** series, we observe that the value of $\omega_0 = 5.5$ eV is simply an average of the excitation energies ω_1 and ω_2 obtained in the two-block model treatment. Furthermore, both model expressions result in the same coupling constant (Eq. (3): 1.33 eV, Eq. (10): 1.31 eV), which confirms the consistency of our approaches. The bridged substituents in the **Cp-OT** series reduce slightly the ω_0 value to 5.61 eV with respect to the pure thiophene series ($\omega_0 = 5.65$ eV). The bridge causes a better planarisation between the thiophene rings and consequently an improved conjugation along the π -system. Indeed, the highest J value of 1.63 eV is obtained from Eq. (3) for the **Cp-OT** series. The parameters for the phenylene-series, **OPP**, show a higher ω_0 value than for the **OT** series (6.16 eV vs. 5.65 eV) in accordance with the higher aromatic character of the latter one. The low coupling constant for the **OPP** series ($J = 1.265$ eV) reflects the loss of planarity between phenylenes due to the steric hindrance caused by the hydrogen-atoms from neighbouring rings. The vinylene units in the mixed thiophene series, **OTV**, show a very high excitation energy ($\omega_2 = 10.82$ eV), acting as barrier for the exciton delocalization. A better coupling between the monomers can be concluded from the high J value in this series (2.69 eV vs. 1.54 eV for **OT**). The improved planarisation of the π -system and consequently the higher J value is also demonstrated in the phenylenevinylene series, **Alkoxy-OPV**, ($J = 2.134$ eV vs. 1.82 for **OPE**) and can be seen as a general effect for the mixed vinylene series. The donating effect of the alkoxy chains on the phenylene moieties is reflected in the lowering of the on-site excitation (4.77 eV vs. 6.164 eV for the **OPP** series). The considerable reduction of the excitation energy ($\Delta\omega = 1.39$ eV) can be explained by the stronger donating character of the alkoxy chains in comparison to the alkyl chains ($\Delta\omega = 0.35$ eV).

Finally, we point out that in all cases where the model with two-building blocks per repeat unit was discussed, with respect to optical absorption Eq. (3) gives also very accurate fits. The fitting parameters for all oligomer series considered by us are summarized in Table 1. For example, the **OTV** and **Alkoxy-OPV** absorption data can be very well fitted within the one-building block model with $\omega_0 = 5.08$ eV, $J = 1.47$ eV ($R^2 = 0.99901$) and $\omega_0 = 4.49$ eV, $J = 0.99$ eV ($R^2 = 0.99902$), respectively. Such a flexibility in choosing

the repeat unit definition is not surprising in the excitonic picture. To establish a link between the two approaches we only have to realize that in the case of fitting Eq. (3) to an oligomer series with two blocks per unit cell we consider the two blocks, thiophene and ethenylene for example, to form a dimer. The state with the higher transition dipole moment of such a dimer serves as an excitation of an effective single block. This can easily be seen by solving the secular equation of a 2 by 2 matrix with diagonal elements given by the excitation energy of each of the two blocks in the repeat unit and off-diagonal elements given by the coupling J . For example, in the **OTV** series, the diagonal elements are 5.65 eV and 10.82 eV with the values for the off-diagonal elements given by -2.69 eV. Solving the corresponding eigenvalue problem we will get the lower eigenvalue as 4.5 eV which is not much different from 5.08 eV resulting from the direct fit of the Eq. (3) to the **OTV** series. The 0.5 eV shift comes from the fact that the **OTV** series consist of an odd number of monomers thus one of the monomers is unpaired and the matching cannot be perfect. As another example we consider the even **OPE** series. From the fit with Eq. (10) we obtained for the diagonal elements, i.e. the excitation energies of the two blocks, $\omega_1 = 5.43$ eV and $\omega_2 = 9.5$ eV and for the off-diagonal element $J = 1.82$ eV. The lower energy in the solution of the dimer problem is 4.74 eV. The **OPE** series fitted with Eq. (3) gives $\omega_0 = 4.70$ eV and $J = 0.72$ eV which perfectly agree with the single ω_0 -value estimated from the dimer. This qualitative interpretation is analogous to the procedure in Ref. [29] where the model parameters of the exciton Hamiltonian were determined from the larger number of energy values obtained from INDO/SCI calculations.

5. Concluding remarks

On the basis of the two model descriptions (with one and two-building blocks per repeat unit) we can conclude that the electronic properties with respect to the position of the optical absorption of the various oligomer series presented are described correctly by both. Moreover, the parameters obtained from the theoretical expressions are not only fitting parameters but have a physical meaning: ω_0 correlates with the excitation energy of the building block in the unit cell and J describes the coupling strength between the building blocks, which is directly related with the excitation delocalization along the oligomer. Analogously, ω_1 and ω_2 are the electronic excitation energies in the model with two-building blocks per repeat unit, and J describes the interaction between the units which in our procedure (for symmetry reasons) was assumed to be the same between all building blocks.

Our study leads to the conclusion that the length-dependent optical effects of conjugated thiophene and benzene-based oligomers can be described and easily understood in the framework of our models based on Frenkel exciton theory. The saturation of the position of the absorption maxima for the longer oligomers can be explained solely by the

size-dependent exciton band width. As discussed in Section 2, the conjugation length is assumed to be equal to the oligomer chain length. In the case of polymers, due to their polydispersity and inherent defects, the experimental energy gap cannot be compared with the value for an infinite oligomer given in our fitting procedure. However, fitting the experimental energy gap for polydisperse systems with our models will result in a good estimation of their mean chain length.

Finally, we showed that the model with two-building blocks per unit cell gives more detailed estimates of the excitation energies of each building block within the repeat unit and for example the influence of attached side groups on the HOMO–LUMO excitation can therefore easily be seen. However, it should be mentioned that despite the large progress in the synthesis of the longer oligomers, the limitation given by the size of the considered series, especially **OPE** and **OPE**, allowed us to estimate the excitation energies of the vinylene and the ethynylene units (i.e. of the two-building blocks in the repeat unit) only under the condition that the excitation of the thiophene or the benzene subunits are known before. A larger number of experimental data would allow for a complete fit without “external support”.

In this paper we have concentrated ourselves on the dependence of the line position of optical absorption, i.e. of a linear optical process, on the size of oligomers and described it in models with one or two (where appropriate) building blocks in the repeat unit. In both models the position of the main absorption line could be described. Future studies of non-linear optical phenomena should result in a series of interesting effects. Furthermore, the oligomers considered in our investigation were monodisperse (with respect to the chain length). The shape and width of the optical absorption line, however, depends heavily on disorder effects, e.g. conformational and orientational disorder, and on various vibrational degrees of freedom. Such more refined investigations will be the subject of future work, which should also include quantum chemical calculations, e.g. along the lines of Ref. [29], in order to get an independent estimate for the model parameters. A long term goal might be to assemble a tool kit, which allows for the construction of model Hamiltonians on the basis of the building blocks and for the derivation of optical properties of oligomers without the use of extensive quantum chemical calculations.

Acknowledgement

The authors gratefully acknowledge the financial support of the Deutsche Forschungsgemeinschaft within SFB 569.

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